

Week 2 — Convexity, LP, and Probability: the Two Reviews

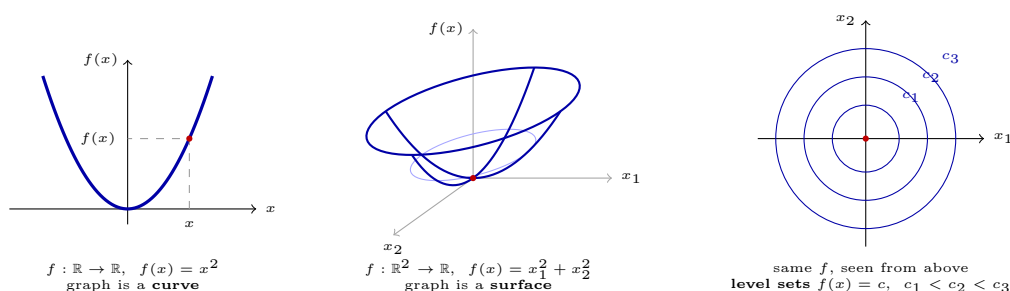
AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Sep 5, 2026

0. Warm-up: functions, vectors, matrices

three objects carry the whole course: a *function*, a *vector*, a *matrix*.

Function $f : A \rightarrow B$ assigns to each input $x \in A$ exactly one output $f(x) \in B$. $A = \text{domain}$, $B = \text{codomain}$.

- $f : \mathbb{R} \rightarrow \mathbb{R}$ one number in, one number out. graph = $\{(x, f(x))\}$ — a curve in the plane
- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ two numbers in, one number out. graph — a surface in 3D; level set $\{x : f(x) = c\}$ — a curve in the plane
- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ an objective is always of this shape: it must return a *single* number, otherwise two decisions cannot be compared.

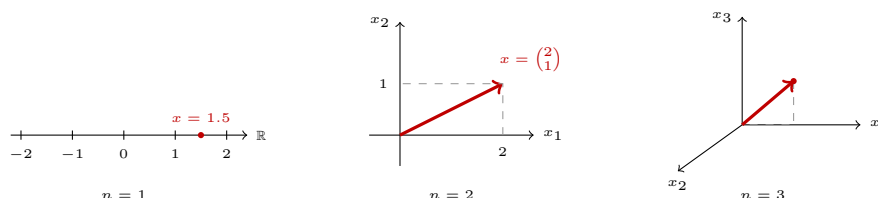


Real space $\mathbb{R}$ = all real numbers — a line. $\mathbb{R}^n$ = lists of n real numbers:

$$x = (x_1, x_2, \dots, x_n), \quad x_i \in \mathbb{R}$$

$n = 1$: line, $n = 2$: plane, $n = 3$: space, $n \geq 4$: same algebra, no picture

“ $x \in \mathbb{R}^n$ ” = one list of n numbers = one point = one arrow from the origin = **one decision**



Vector by convention a *column*: $x \in \mathbb{R}^n$ is $n \times 1$. transpose $x^\top$ is the row, $1 \times n$.

- **inner product** of $c, x \in \mathbb{R}^n$: $c^\top x = c_1 x_1 + \dots + c_n x_n = \sum_{i=1}^n c_i x_i$
as a function: $f(x) = c^\top x, f : \mathbb{R}^n \rightarrow \mathbb{R}$ — fix c , feed it x , get one number
Section 3’s objective: c = prices, x = amounts, $f(x) = c^\top x$ = value of the cargo

- **2-norm** of $x \in \mathbb{R}^n$: $\|x\|_2 = \sqrt{x^\top x} = \sqrt{x_1^2 + \dots + x_n^2}$
as a function: $f(x) = \|x\|_2, f : \mathbb{R}^n \rightarrow \mathbb{R}$ — again one number out

Matrix a matrix is a *stack of row vectors*:

$$A = \begin{pmatrix} - & a_1^\top & - \\ - & a_2^\top & - \\ & \vdots & \\ - & a_m^\top & - \end{pmatrix} \in \mathbb{R}^{m \times n}, \quad a_1, a_2, \dots, a_m \in \mathbb{R}^n$$

m = number of rows, n = length of each row.

Example $m = 3, n = 4$:

$$A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 3 & 1 & 4 & 0 \\ 0 & 2 & 1 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 4}, \quad a_1 = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 1 \\ 4 \\ 0 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 5 \end{pmatrix} \in \mathbb{R}^4$$

- **matrix $\times$ vector**: for $x \in \mathbb{R}^n$, $Ax \in \mathbb{R}^m$ with $(Ax)_i = a_i^\top x$ — one inner product per row

$$x = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^4: \quad Ax = \begin{pmatrix} a_1^\top x \\ a_2^\top x \\ a_3^\top x \end{pmatrix} = \begin{pmatrix} 1 + 6 + 0 + 0 \\ 3 + 2 + 4 + 0 \\ 0 + 4 + 1 + 0 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \\ 5 \end{pmatrix} \in \mathbb{R}^3$$

as a function: $f(x) = Ax$, $f: \mathbb{R}^4 \rightarrow \mathbb{R}^3$

— takes a 4-dimensional vector, returns a 3-dimensional vector; not a single number (a *scalar*)

in general $f(x) = Ax$, $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

- **identity matrix** $I \in \mathbb{R}^{n \times n}$: 1's on the diagonal, 0's elsewhere. its rows are $e_1^\top, \dots, e_n^\top$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}, \quad (Ix)_i = e_i^\top x = x_i, \text{ so } Ix = x \text{ for every } x$$

- **transpose** $A^\top \in \mathbb{R}^{n \times m}$: rows become columns. A is **symmetric** if $A = A^\top$ (so $m = n$)

Quadratic function

- one variable: $f(x) = px^2 + qx + r$ — a parabola; opens upward iff $p \geq 0$
- n variables: $f(x) = x^\top Qx + q^\top x + r$, $Q \in \mathbb{R}^{n \times n}$ symmetric — still $\mathbb{R}^n \rightarrow \mathbb{R}$, one number out

$$n = 2: \quad x^\top Qx = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{12} & Q_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = Q_{11}x_1^2 + 2Q_{12}x_1x_2 + Q_{22}x_2^2$$

$Q = I$: $x^\top x = x_1^2 + x_2^2$ — the round bowl drawn above. “opens upward” becomes $Q \succeq 0$ (Sections 5, 6).

1. General optimization problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & x \in X \end{array}$$

1. what do we control? **decision variables** x
2. what do we want? **objective function** f
3. what limits us? constraints $\rightarrow$ **feasible set** X

$$X = \{x \in \mathbb{R}^n : g_i(x) \leq 0 \ \forall i \in [m]\}$$

in plain English: x belongs to the n -dimensional real space $\mathbb{R}^n$, **and** it satisfies all m constraints

$$g_1(x) \leq 0, \quad g_2(x) \leq 0, \quad \dots, \quad g_m(x) \leq 0$$

read the symbols: $\in$ = “is in”, $\forall$ = “for all”, $[m] := \{1, \dots, m\}$, s.t. = “subject to”

— so “ $\forall i \in [m]$ ” packs that whole list of m constraints into one stroke

$$h(x) = 0 \iff h \leq 0 \text{ and } -h \leq 0$$

takeaway how we define $f(x)$ and X determines the program class. some problems are easy to solve — but some are not.

2. One problem, three languages

	OR	ML	control
decision	x	θ	u
uncertain data	ξ	[x = input, y = output] state x perturbed by noise w	

keep an eye on that ξ — *uncertain data* is exactly what this course is about. it returns in Section 7.

KEY optimization is not a topic inside OR — it is the shared language of OR, ML, control.

watch Brunton, *Optimization: A Bootcamp for Machine Learning, Inverse Problems, and Control* — 22 min, supplementary.

https://www.youtube.com/watch?v=1PBPbGmw1_4

3. Linear programs — and why they are easy

$$\begin{aligned} \max_x \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

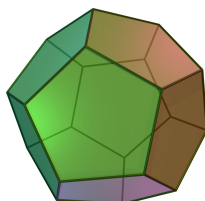
$$c \in \mathbb{R}^n, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m$$

$$\text{row } i: \quad g_i(x) = a_i^\top x - b_i \leq 0 \iff a_i^\top x \leq b_i$$

$$\text{stack rows} \Rightarrow A \quad \text{so} \quad a_i^\top x \leq b_i \quad \forall i \iff Ax \leq b$$

in many cases variables are nonnegative ($x \geq 0$) — encoded in $Ax \leq b$ by the rows $-x_j \leq 0$

$P = \{x : Ax \leq b\}$ is called a **polyhedron**



Polyhedron in $\mathbb{R}^3$. (image: Wikimedia Commons)

Convexity

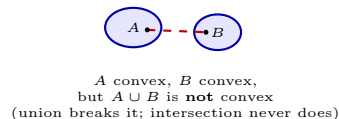
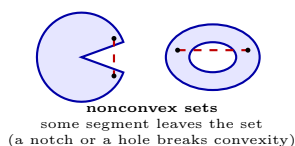
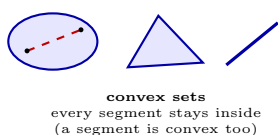
- **Convex set:** $S \subseteq \mathbb{R}^n$ is convex if $\lambda x + (1 - \lambda)y \in S$ for all $x, y \in S$ and all $\lambda \in [0, 1]$
- **Convex function:** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in \mathbb{R}^n$ and all $\lambda \in [0, 1]$
strictly convex if strict for $x \neq y$, $\lambda \in (0, 1)$
- **Convex problem:** $\min\{f(x) : x \in S\}$ with S convex *and* f convex

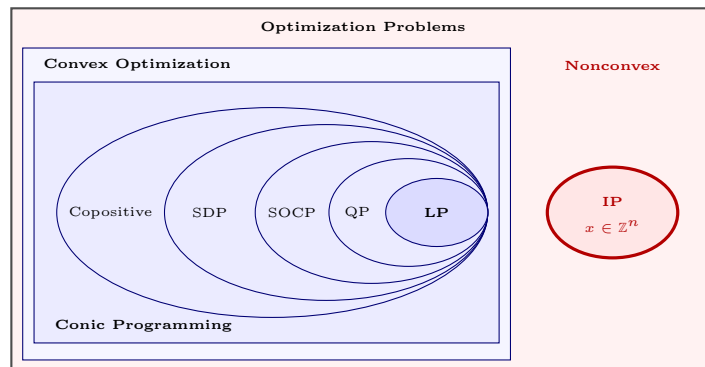
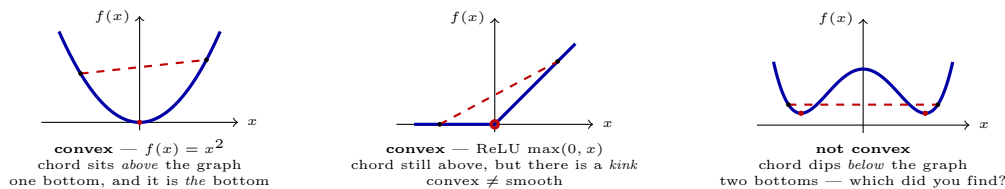
Prop. in a convex problem, every local min is a global min.

pf sketch: if $f(y) < f(x^*)$, then $f(\lambda y + (1 - \lambda)x^*) \leq \lambda f(y) + (1 - \lambda)f(x^*) < f(x^*)$
for all $\lambda \in (0, 1]$ — so better points sit arbitrarily close to x^* .

Examples

convex	not convex
least squares $\ Ax - b\ _2^2$ over unconstrained $\mathcal{X}$ — one bowl	neural-net training loss — many dips
linear $c^\top x$ over P — our LP	$\in \mathbb{Z}$: feasible set turns discrete! — OL7014's turf





LP $\subset$ QP $\subset$ SOCP $\subset$ SDP $\subset$ Copositive $\subset$ Conic $\subset$ Convex, but IP $\not\subset$ Convex
 (QP = quadratic, SOCP = second-order cone, SDP = semidefinite)
 local = global holds everywhere inside the convex box — **this course lives inside it**

takeaway an LP is a convex problem: P convex (intersection of half-spaces) + $c^\top x$ linear, hence a convex function.

4. First LP problem

x = crates of ammo (dense), y = crates of rations (bulky)

$$\begin{aligned} \max \quad & 4x + 5y \\ \text{s.t.} \quad & x + 3y \leq 10 \quad (\text{volume}) \\ & 3x + y \leq 10 \quad (\text{weight}) \\ & x, y \geq 0 \end{aligned}$$

now put it in standard vector form:

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & 3 \\ 3 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 10 \\ 10 \\ 0 \\ 0 \end{pmatrix} \quad (\text{last two rows are } x, y \geq 0)$$

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{aligned} \quad \text{— Section 3, now with numbers in it}$$

vertices: $(0, 0)$, $(\frac{10}{3}, 0)$, $(2.5, 2.5)$, $(0, \frac{10}{3})$

slide $4x + 5y = z$ up until it leaves $\Rightarrow \boxed{(x^*, y^*) = (2.5, 2.5), \quad z_{\text{LP}} = 22.5}$

KEY 1. an optimum sits at a vertex.

2. fractional answers are normal — nothing forces an LP onto whole numbers.

demo LP-Graphical-Demo.exe — animation slides the line; every vertex labeled with $c^\top v$.

5. One rung up: quadratic programs

one ring out on Section 3's map: keep the polyhedron, let the objective curve.

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^\top Q x + c^\top x \\ \text{s.t.} \quad & \mathbf{A}x \leq \mathbf{b} \end{aligned}$$

convex exactly when $Q \succeq 0$ — “the bowl opens upward.” ($Q = 0$: flat bowl, back to LP.)

two QPs we already know: • least squares $\min \|Ax - b\|_2^2$ — a QP with *no* constraints at all • Markowitz (week 3’s notebook, Section 8): $\min x^\top \Sigma x$ s.t. $\mu^\top x \geq r$, $\mathbf{1}^\top x = 1$, $x \geq 0$

Section 4’s cargo, one more request the field unit asks for the exact mix $(\bar{x}, \bar{y}) = (4, 2)$ — infeasible! (weight: $3 \cdot 4 + 2 = 14 > 10$.) so get *as close as possible*:

scalar form

$$\begin{array}{ll} \min & (x - 4)^2 + (y - 2)^2 \\ \text{s.t.} & x + 3y \leq 10 \\ & 3x + y \leq 10 \\ & x, y \geq 0 \end{array}$$

vector form — with numbers in it

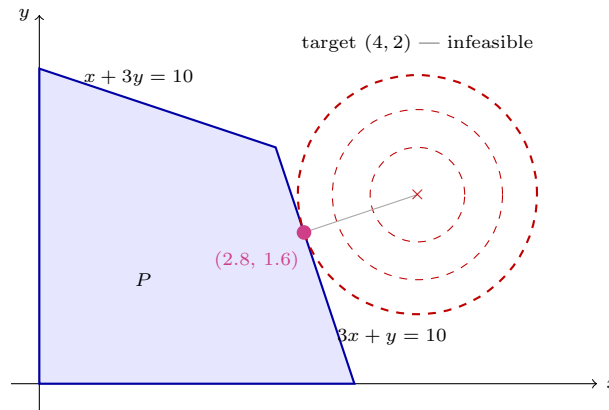
$$\begin{array}{ll} \min & \underbrace{\frac{1}{2} \mathbf{x}^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{x}}_{Q=2I} + \underbrace{\begin{pmatrix} -8 & -4 \end{pmatrix} \mathbf{x}}_{c^\top} + 20 \\ \text{s.t.} & A\mathbf{x} \leq b \quad \text{— same } A, b \text{ as Section 4} \end{array}$$

check: expand $(x - 4)^2 + (y - 2)^2 = x^2 + y^2 - 8x - 4y + 20$; the constant 20 moves no minimizer.

level sets are *circles* around the target ($Q = 2I$: a perfectly round bowl. general $Q \succeq 0$ stretches circles into ellipses.)

solve it graphically: inflate the circle until it first touches P — it lands *tangent to the middle of the weight wall*. drop a perpendicular from the target onto $3x + y = 10$:

$$\boxed{(x^*, y^*) = (2.8, 1.6), \quad f^* = 1.6} \quad \text{— not a vertex.}$$



constrained QP, graphically
inflate the level sets until they first touch P : tangent to an edge — not at a vertex

in the control column of Section 2’s table, the same move — quadratic cost over linear structure — is called *LQR*. same rung, third language.

KEY 1. curved objective $\Rightarrow$ the optimum can leave the vertices: interior (target feasible), edge (here), or vertex.
2. it stops wherever the lowest level set first touches P — LP’s flat level sets slide to a corner.

demo QP-Graphical-Demo.exe — two panels: inflate the circle in 2D, watch the bowl in 3D; drag the target — interior, edge, vertex.

demo week03_markowitz.ipynb — week 3’s Colab — Markowitz as a constrained QP; solve it, then trace the frontier.

read Caron, *Quadratic Programming in Python* — same picture, plus solver code.

<https://scaron.info/blog/quadratic-programming-in-python.html>

takeaway a QP is LP’s polyhedron plus a curved bowl: still convex, still local = global — but the vertex story is over.

6. Convexity-preserving operations

(Boyd & Vandenberghe, *Convex Optimization*, Part I)

$$\text{dom}(f) = \{x \in \mathbb{R}^n : f(x) < +\infty\}$$

Examples of convex functions

<i>univariate</i>	affine function	$ax + b$
	quadratic function	$px^2 + qx + r \quad \text{if } p \geq 0$
	power function	$x^\alpha \quad (\alpha \geq 1 \text{ or } \alpha \leq 0)$
	negative logarithm	$-\log x$
<i>multivariate</i>	affine function	$a^\top x + b$
	quadratic function	$x^\top Px + x^\top q + r \quad \text{if } P \succeq 0$
	norm	$\ x\ _p \quad \text{for } p \geq 1$
$P \succeq 0$ (<i>positive semidefinite</i>) $\iff x^\top Px \geq 0 \quad \forall x \in \mathbb{R}^n$		

Convexity-preserving operations

1) Composition with affine functions:

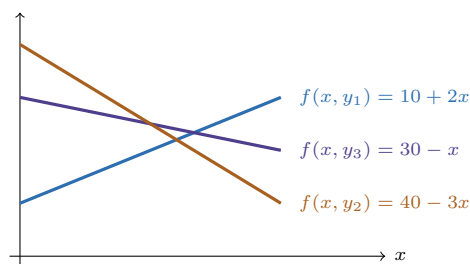
$f(x)$ is convex in x , then $g(x) = f(Ax + b)$ is also convex in x .

2) Nonnegative weighted sum:

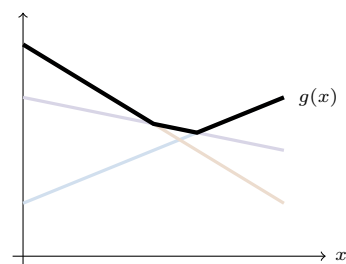
$f_1, f_2, \dots, f_k$ are convex in x , then $g(x) = w_1 f_1(x) + \dots + w_k f_k(x)$ is also convex in x for $w_1, w_2, \dots, w_k \geq 0$.

3) Pointwise supremum:

If $f(x, y)$ is convex in x for every fixed $y \in Y$, then $g(x) = \sup_{y \in Y} f(x, y)$ is also convex in x .



three affine functions of x
(one for each $y \in Y = \{y_1, y_2, y_3\}$)



$g(x) = \max\{f(x, y_1), f(x, y_2), f(x, y_3)\}$
convex, piecewise linear — kinks where the leader changes

4) Partial minimization:

If $f(x, y)$ is *jointly convex* in x and y , and C is a convex set, then $g(x) = \inf \{ f(x, y) : y \in C \}$ is convex in x .

$$\text{jointly convex:} \quad f \left[\lambda \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + (1 - \lambda) \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right] \leq \lambda f \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + (1 - \lambda) f \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$$\forall \lambda \in [0, 1], \quad \forall \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$f(x, y) = xy$ is convex in x and in y separately, but *not* jointly convex.

Back to what we have already used

least squares $\|Ax - b\|_2^2$ $\|\cdot\|_2^2$ convex, then (1)

piecewise linear $\max_i (a_i^\top x + b_i)$ affine convex, then (3)

Markowitz $x^\top \Sigma x$ quadratic with $\Sigma \succeq 0$ ($= \|Lx\|_2^2$ for $\Sigma = L^\top L$: (1) again)

polyhedron $P = \{x : Ax \leq b\}$ intersection of m half-spaces — intersection of convex sets is convex

takeaway convexity is compositional: certify the pieces, then check the operation. the risk measures (CVaR, week 4) and worst-case objectives ahead are built exactly this way.

7. Probability review

back to Section 2's table: the OR column had **uncertain data** ξ . today, the vocabulary — textbook notation (X, f) first, this course's notation $(\tilde{\xi})$ at the end.

Random variable

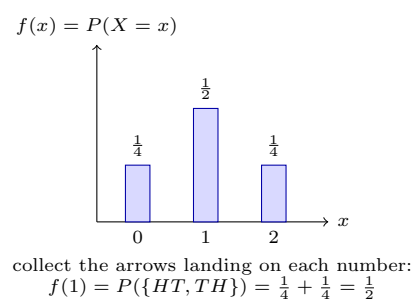
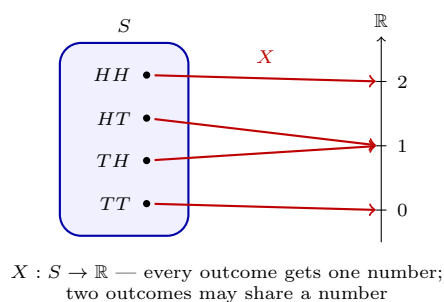
- **random experiment:** its outcome is not known in advance
- **sample space** S = the set of all possible outcomes

Def (*random variable*) a function that assigns a real number to each outcome: $X : S \rightarrow \mathbb{R}$

capital X = the random variable, lowercase x = a value it takes

toss a coin twice: $S = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$

X = number of heads: $X(HH) = 2$, $X(HT) = 1$, $X(TH) = 1$, $X(TT) = 0$



- **discrete** X : countably many values — number of defective parts, number of arrivals, demand
- **continuous** X : values fill an interval — length, time, temperature, a return

Discrete random variables

Def (*probability mass function, PMF*) $f(x) = P(X = x)$, $f : \mathbb{R} \rightarrow [0, 1]$

(1) $f(x) \geq 0$ (2) $\sum_x f(x) = 1$ $P(X \in A) = \sum_{x \in A} f(x)$

two coins: $f(0) = \frac{1}{4}$, $f(1) = \frac{1}{2}$, $f(2) = \frac{1}{4}$, $P(X \geq 1) = f(1) + f(2) = \frac{3}{4}$

Def (*mean, variance*)

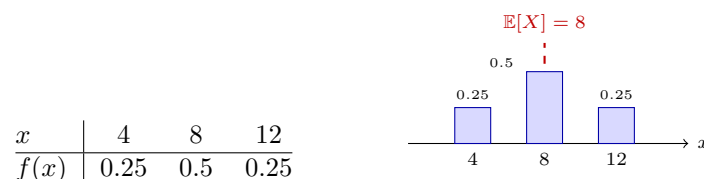
mean: $\mu = \mathbb{E}[X] = \sum_x x f(x)$

variance: $\sigma^2 = \text{Var}(X) = \sum_x (x - \mu)^2 f(x) = \mathbb{E}[X^2] - \mu^2$, standard deviation $\sigma = \sqrt{\text{Var}(X)}$

function of X : $\mathbb{E}[h(X)] = \sum_x h(x) f(x)$ $\leftarrow$ law of the unconscious statistician

$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b$, $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Example 1 demand X for a spare part next month:



$$\mathbb{E}[X] = 0.25 \cdot 4 + 0.5 \cdot 8 + 0.25 \cdot 12 = 8$$

$$\text{Var}(X) = 0.25 \cdot (4 - 8)^2 + 0.5 \cdot (8 - 8)^2 + 0.25 \cdot (12 - 8)^2 = 8, \quad \sigma = 2\sqrt{2} \approx 2.83$$

Continuous random variables

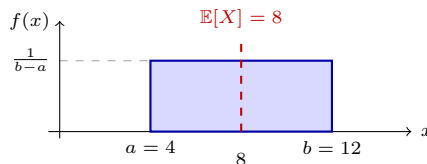
Def (*probability density function, PDF*) $f : \mathbb{R} \rightarrow \mathbb{R}_+$ with

- (1) $f(x) \geq 0$ (2) $\int_{-\infty}^{\infty} f(x) dx = 1$ (3) $P(a \leq X \leq b) = \int_a^b f(x) dx$ = area under f from a to b
 $f(x)$ is not a probability: $P(X = x) = 0$ for every single x , and $f(x) > 1$ is allowed

Def (*mean, variance*) same as discrete — the sums become integrals:

$$\mu = \mathbb{E}[X] = \int x f(x) dx, \quad \sigma^2 = \text{Var}(X) = \int (x - \mu)^2 f(x) dx = \mathbb{E}[X^2] - \mu^2, \quad \mathbb{E}[h(X)] = \int h(x) f(x) dx$$

Example 2 $X \sim \text{Uniform}[a, b]$ — $f(x) = \frac{1}{b-a}$ on $[a, b]$, 0 outside



mean:

$$\mathbb{E}[X] = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

second moment — $h(x) = x^2$:

$$\mathbb{E}[X^2] = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

variance — subtract, no new integral:

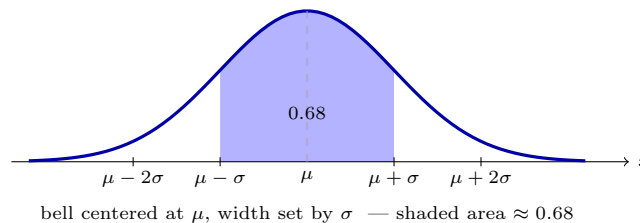
$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

with $a = 4, b = 12$: $\mathbb{E}[X] = 8, \quad \text{Var}(X) = \frac{64}{12} = \frac{16}{3}, \quad \sigma = \frac{4}{\sqrt{3}} \approx 2.31$

Normal distribution $X \sim \mathcal{N}(\mu, \sigma^2)$: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2$$

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68, \quad P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$



Two random variables

Def (*joint PMF*) $f_{XY}(x, y) = P(X = x, Y = y), \quad f_{XY} \geq 0, \quad \sum_x \sum_y f_{XY}(x, y) = 1$

Def (*marginal PMF*) $f_X(x) = \sum_y f_{XY}(x, y), \quad f_Y(y) = \sum_x f_{XY}(x, y)$

— row sums and column sums, written in the *margins* of the table

Example 3 two assets, returns tomorrow: X, Y each either down (-0.10) or up ($+0.20$)

$f_{XY}(x, y)$	$y = -0.10$	$y = 0.20$	$f_X(x)$
$x = -0.10$	0.3	0.2	0.5
$x = 0.20$	0.2	0.3	0.5
$f_Y(y)$	0.5	0.5	1

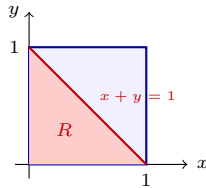
each asset alone: down or up w.p. $\frac{1}{2}$. together: same direction w.p. $0.3 + 0.3 = 0.6$

Def (joint PDF) $f_{XY} : \mathbb{R}^2 \rightarrow \mathbb{R}_+$, $\iint f_{XY}(x, y) dx dy = 1$, $P((X, Y) \in R) = \iint_R f_{XY}(x, y) dx dy$

— the **volume** under f_{XY} above the region R

marginal PDF: $f_X(x) = \int f_{XY}(x, y) dy$ — the sum becomes an integral

Example 4 (X, Y) uniform on the unit square: $f_{XY}(x, y) = 1$ on $[0, 1]^2$, 0 outside



$$P(X + Y \leq 1) = \iint_R 1 dx dy = \text{area of the triangle } R = \frac{1}{2}$$

$$\text{marginal: } f_X(x) = \int_0^1 1 dy = 1 \text{ on } [0, 1] \text{ — } X \text{ alone is Uniform}[0, 1]$$

Def (independence) X, Y are independent if $f_{XY}(x, y) = f_X(x) f_Y(y) \quad \forall x, y$ (PMF or PDF)

Example 3: $f_{XY}(-0.10, -0.10) = 0.3 \neq 0.5 \cdot 0.5$ — **not** independent

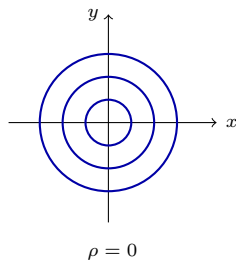
Example 4: $f_{XY}(x, y) = 1 = 1 \cdot 1$ — independent

Def (covariance, correlation)

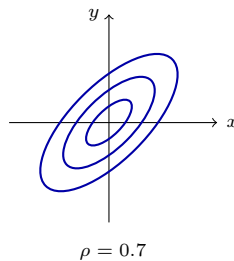
$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y, \quad \text{Cov}(X, X) = \text{Var}(X)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho_{XY} \leq 1$$

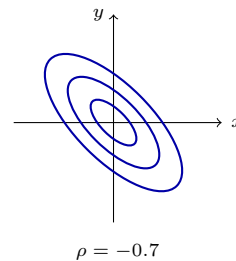
$\rho > 0$: move together, $\rho < 0$: move opposite, independent $\Rightarrow \rho = 0$



$$\rho = 0$$



$$\rho = 0.7$$



$$\rho = -0.7$$

contours of the joint PDF of two normal random variables, $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$

Example 3, continued (use the margins for μ and σ , all four cells for the covariance)

$$\mu_X = 0.5 \cdot (-0.10) + 0.5 \cdot 0.20 = 0.05, \quad \mu_Y = 0.05$$

$$\text{Var}(X) = 0.5 \cdot (-0.15)^2 + 0.5 \cdot (0.15)^2 = 0.0225, \quad \sigma_X = \sigma_Y = 0.15$$

$$\begin{aligned} \text{Cov}(X, Y) &= 0.3(-0.15)(-0.15) + 0.2(-0.15)(0.15) + 0.2(0.15)(-0.15) + 0.3(0.15)(0.15) \\ &= (0.3 - 0.2 - 0.2 + 0.3) \cdot 0.0225 = 0.0045 \end{aligned}$$

$$\rho_{XY} = \frac{0.0045}{0.15 \cdot 0.15} = 0.2 \quad \text{— the two assets tend to move together}$$

Linear combinations

$$\mathbb{E}[aX + bY] = a\mu_X + b\mu_Y \quad \text{— always, no independence needed}$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

n random variables $X = (X_1, \dots, X_n)$:

mean vector $\mu = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_n])^\top \in \mathbb{R}^n$
 covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, $\Sigma_{ij} = \text{Cov}(X_i, X_j)$; diagonal $\Sigma_{ii} = \text{Var}(X_i)$; symmetric; always $\Sigma \succeq 0$
 for fixed $c \in \mathbb{R}^n$: $\mathbb{E}[c^\top X] = c^\top \mu$, $\text{Var}(c^\top X) = c^\top \Sigma c$
 $n = 2$, $c = (a, b)$: $c^\top \Sigma c = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \text{Var}(aX + bY)$ again
 Example 3: $\mu = \begin{pmatrix} 0.05 \\ 0.05 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 0.0225 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix}$

In this course's notation the decision is already x , so the uncertain data gets its own letter:

	textbook	this course
random variable	X	$\tilde{\xi}$, with distribution $\mathbb{P}$: $\tilde{\xi} \sim \mathbb{P}$
a value of it	x	ξ (a <i>realization</i>)
probability	$P(\cdot)$	$\mathbb{P}(\cdot)$
set of possible values	range of X	support Ξ (smallest closed set with $\mathbb{P}(\tilde{\xi} \in \Xi) = 1$)
discrete	values x , PMF $f(x)$	scenarios $\xi_1, \dots, \xi_S$ with probabilities $p_1, \dots, p_S$
continuous	PDF $f(x)$	density $p(\xi)$
mean, discrete	$\sum_x x f(x)$	$\mathbb{E}[\tilde{\xi}] = \sum_{s \in [S]} p_s \xi_s$
mean, continuous	$\int x f(x) dx$	$\mathbb{E}[\tilde{\xi}] = \int_{\Xi} \xi p(\xi) d\xi$
function of it	$\mathbb{E}[h(X)]$	$\mathbb{E}[f(\tilde{\xi})]$ (f = any function, e.g. a cost — not a PDF)
random vector	$X \in \mathbb{R}^k$, μ , Σ	$\tilde{\xi} \in \mathbb{R}^k$, μ , Σ

The bridge to optimization data random $\Rightarrow$ cost random: $f(x, \tilde{\xi})$ is a random variable, $\mathbb{E}[f(x, \tilde{\xi})]$ is a number.

$$\min_x \mathbb{E}[f(x, \tilde{\xi})]$$

takeaway every paradigm in this course differs in one place only: *how it collapses the randomness to a number.*

8. Running example: the Markowitz portfolio

a budget of 1, n assets. decide the split today; the returns are revealed tomorrow.

- **decision:** $x \in \mathbb{R}^n$, x_j = share of the budget in asset j
- **uncertain data:** $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n) \in \mathbb{R}^n$, $\tilde{\xi}_j$ = return of asset j — a random vector with mean μ , covariance matrix Σ
- **feasible set:** $X = \{x \in \mathbb{R}^n : \mathbf{1}^\top x = 1, x \geq 0\}$ ($\mathbf{1}$ = all-ones vector, $\mathbf{1}^\top x = x_1 + \dots + x_n$)

Portfolio return $\tilde{\xi}^\top x = x_1 \tilde{\xi}_1 + \dots + x_n \tilde{\xi}_n$ — a random variable

as a function: fix x , $f(\xi) = \xi^\top x$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ — Section 0's inner product, fed a random input

loss: $f(x, \tilde{\xi}) = -\tilde{\xi}^\top x$ (a gain is a negative loss)

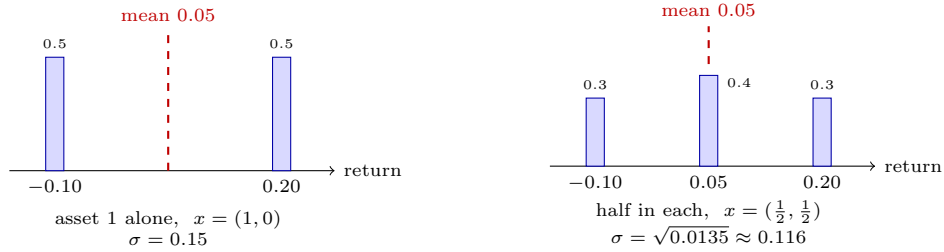
Section 7's linear combinations with $c = x$: $\mathbb{E}[\tilde{\xi}^\top x] = \mu^\top x$, $\text{Var}(\tilde{\xi}^\top x) = x^\top \Sigma x$

Check on Example 3 $(\tilde{\xi}_1, \tilde{\xi}_2) = \text{Example 3's } (X, Y)$, half in each: $x = (\frac{1}{2}, \frac{1}{2})$

ξ_1	ξ_2	$\frac{1}{2}\xi_1 + \frac{1}{2}\xi_2$	probability	$\Rightarrow$	PMF of $\tilde{\xi}^\top x$:				
-0.10	-0.10	-0.10	0.3			value	-0.10	0.05	0.20
-0.10	0.20	0.05	0.2			probability	0.3	0.4	0.3
0.20	-0.10	0.05	0.2						
0.20	0.20	0.20	0.3						

from the PMF: $\mathbb{E}[\tilde{\xi}^\top x] = 0.3(-0.10) + 0.4(0.05) + 0.3(0.20) = 0.05$
 $\text{Var}(\tilde{\xi}^\top x) = 0.3(-0.15)^2 + 0.4(0)^2 + 0.3(0.15)^2 = 0.0135$

from μ and Σ : $\mu^\top x = \frac{1}{2}(0.05) + \frac{1}{2}(0.05) = 0.05 \checkmark$
 $x^\top \Sigma x = \frac{1}{4}(0.0225) + \frac{1}{4}(0.0225) + 2 \cdot \frac{1}{4}(0.0045) = 0.0135 \checkmark$



same mean, less spread: **diversification**. it comes from $\rho < 1$ — with $\rho = 1$, $x^\top \Sigma x$ would stay 0.0225.

The model (Markowitz, 1952) — least risk, subject to an expected-return target r :

$$\begin{aligned} \min_x \quad & x^\top \Sigma x && \text{variance of the return (risk)} \\ \text{s.t.} \quad & \mu^\top x \geq r && \text{expected return meets the target} \\ & \mathbf{1}^\top x = 1, \quad x \geq 0 \quad x \in X \end{aligned}$$

Section 5's QP with $Q = 2\Sigma$, $c = 0$; convex because $\Sigma \succeq 0$ (Section 6).

in the language of the bridge: the loss $-\tilde{\xi}^\top x$ is random; Markowitz collapses it with two numbers — its variance (objective) and its mean (constraint).

Data — the coding session $n = 3$, $r = 0.10$

asset j	expected return μ_j	variance Σ_{jj}	$\sigma_j = \sqrt{\Sigma_{jj}}$
bond	0.06	0.010	0.10
tech	0.10	0.040	0.20
energy	0.15	0.090	0.30

$$x = \begin{pmatrix} x_{\text{bond}} \\ x_{\text{tech}} \\ x_{\text{energy}} \end{pmatrix}, \quad \mu = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0 \\ 0.001 & 0.040 & 0.015 \\ 0 & 0.015 & 0.090 \end{pmatrix}$$

covariances: bond–tech 0.001, tech–energy 0.015, bond–energy 0

written out term by term — exactly the lines typed in the code:

$$\begin{aligned} \min \quad & 0.010 x_{\text{bond}}^2 + 0.040 x_{\text{tech}}^2 + 0.090 x_{\text{energy}}^2 + 2(0.001) x_{\text{bond}} x_{\text{tech}} + 2(0.015) x_{\text{tech}} x_{\text{energy}} \\ \text{s.t.} \quad & 0.06 x_{\text{bond}} + 0.10 x_{\text{tech}} + 0.15 x_{\text{energy}} \geq 0.10 \\ & x_{\text{bond}} + x_{\text{tech}} + x_{\text{energy}} \leq 1 \\ & x_{\text{bond}}, x_{\text{tech}}, x_{\text{energy}} \geq 0 \end{aligned}$$

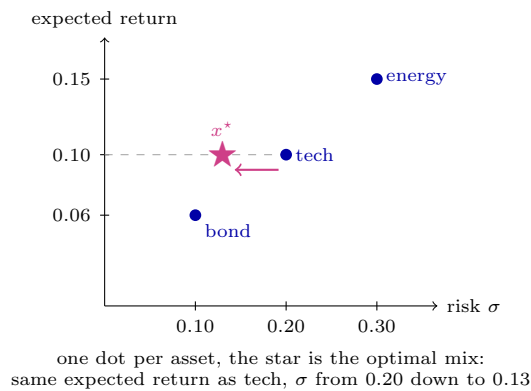
budget “ ≤ 1 ”: leftover sits in cash earning 0 — never helps, so the sum is 1 at the optimum

Solution

$$\boxed{x^* \approx (0.39, 0.29, 0.31), \quad x^{*\top} \Sigma x^* \approx 0.0169, \quad \sigma \approx 0.130} \quad \mu^\top x^* = 0.10 \text{ — target met exactly}$$

all in tech: same expected return 0.10, but $\sigma = 0.20$. the mix cuts σ by about a third.

every asset gets a share — not a vertex (Section 5).



demo `week03_markowitz.ipynb` — Colab — build this model live from the skeleton, then compare with the completed cell.

demo `demos/markowitz_portfolio.py` — the same model as a plain script: prints the model, solves it, plots the weights x^* .

takeaway the running example: the loss $-\tilde{\xi}^\top x$ stays; each later model changes only how it is collapsed to a number — today variance, next CVaR and VaR (week 4).

next lecture duality — every LP comes with a shadow LP: prices, certificates, and why $\max = \min$; then the competing paradigms for OUU.